

Trig Integration

Section 8.3

1-30-26

A) Identity Reminder

$$\cos(2x) = 2\cos^2 x - 1$$

$$2\cos^2 x = 1 + \cos(2x)$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

$$\cos(2x) = 1 - 2\sin^2 x$$

$$2\sin^2 x = 1 - \cos(2x)$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

Ex: (#30)

$$\int \cos^2 \frac{\pi x}{2} dx = \frac{1}{2} \int 1 + \cos(\pi x) dx$$

$$\frac{1}{2} \left(x + \frac{1}{\pi} \sin(\pi x) \right) + C$$

Ex: (#28)

$$\int \sin^3 x dx = \int \sin^2 x (\sin x) dx$$

$$= \int (1 - \cos^2 x) (\sin x) dx$$

$$= \int \sin x - \sin x \cos^2 x dx$$

$$= -\cos x + \int u^2 du \quad \begin{array}{l} u = \cos x \\ du = -\sin x \end{array}$$

$$= \boxed{-\cos x - \frac{1}{3}(\sin x)^3 + C}$$

Ex: (#31)

$$\int \tan x \sec^6 x dx =$$

$$\int \sec^5 x (\tan x \sec x) dx =$$

$$u = \sec x$$

$$du = \sec x \tan x$$

$$= \int u^5 du = \frac{u^6}{6} + C$$

$$= \boxed{\frac{\sec^6 x}{6} + C}$$

Ex: (#37)

$$\int_0^{\frac{\pi}{4}} \tan^3 x dx =$$

$$\int_0^{\frac{\pi}{4}} (\sec^2 x - 1) \tan x dx =$$

$$= \int_0^{\frac{\pi}{4}} \tan x \sec^2 x - \tan x$$

$$= \int_0^{\frac{\pi}{4}} \sec x (\tan x \sec x) - \int_0^{\frac{\pi}{4}} \tan x$$

$$u = \sec x$$

$$du = \sec x \tan x$$

$$= \int_1^{\sqrt{2}} u du - \int_0^{\frac{\pi}{4}} \tan x$$

$$= \frac{\sec^2 x}{2} + \ln|\cos x| \Big|_0^{\frac{\pi}{4}}$$

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